

Cyber Security Dashboard Report

1. Objective

The objective of this project is to develop an interactive Cyber Security Dashboard that provides a centralized view of threat monitoring, attack analysis, and vulnerability management. The dashboard helps security teams monitor security incidents, analyze attack trends, track vulnerability remediation progress, and support data-driven decision-making.

2. Overview

Organizations face various cybersecurity challenges, including malware attacks, phishing attempts, ransomware incidents, and system vulnerabilities. Security teams require a centralized platform to monitor these activities efficiently.

This dashboard was developed using Power BI to provide real-time visibility into threat alerts, attack activities, and vulnerability patch status. Through interactive visualizations and key performance indicators, users can quickly identify risks, evaluate security performance, and prioritize remediation efforts.

3. Dataset Description

The project utilizes three datasets:

Threat Alerts Dataset

Contains information related to security alerts generated within the organization.

Key Fields:

- Alert ID
- Date
- Severity
- Alert Type
- Department

Blocked Attacks Dataset

Contains information about attack attempts detected and blocked by security controls.

Key Fields:

- Attack ID
- Attack Type
- Country
- Date
- Blocked Status

Vulnerability Status Dataset

Contains vulnerability and patch management information.

Key Fields:

- Vulnerability ID
- Department
- Severity
- Patch Status
- Date

4. Dashboard KPIs

The dashboard includes the following key performance indicators:

Total Threat Alerts- Displays the total number of threat alerts generated during the selected period.

Critical Alerts- Displays the number of critical severity alerts requiring immediate attention.

Total Attacks- Displays the total number of attack attempts detected.

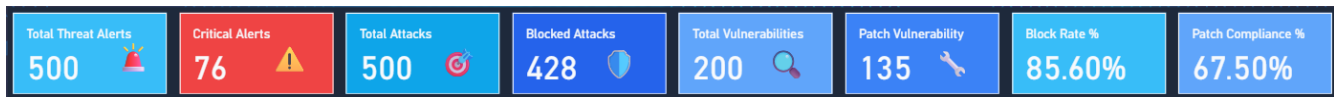
Blocked Attacks- Displays the number of successfully blocked attacks.

Total Vulnerabilities- Displays the total identified vulnerabilities.

Patched Vulnerabilities- Displays the number of vulnerabilities that have been remediated.

Block Rate Percentage- Measures the effectiveness of security controls by calculating the percentage of blocked attacks.

Patch Compliance Percentage- Measures the percentage of vulnerabilities that have been patched.

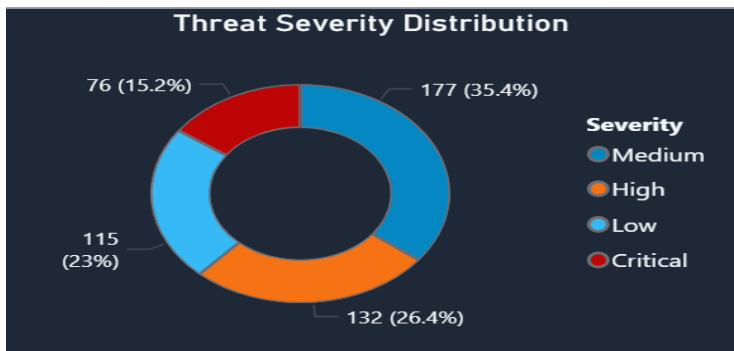


5. Dashboard Visualizations



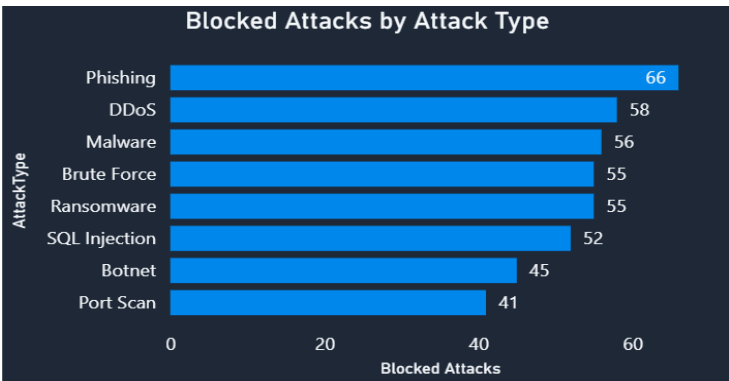
Threat Alerts Trend

A line chart showing changes in threat alert volumes over time. This visual helps identify unusual spikes in security activity and supports proactive monitoring.



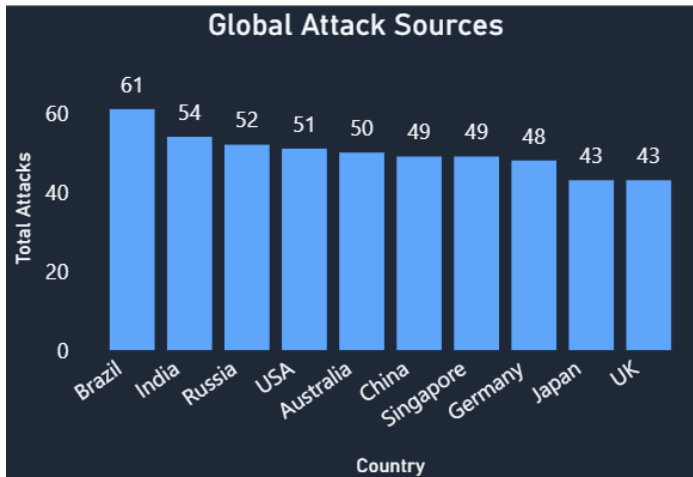
Threat Severity Distribution

A donut chart categorizing alerts into Low, Medium, High, and Critical severity levels. This helps prioritize incident response activities based on risk.



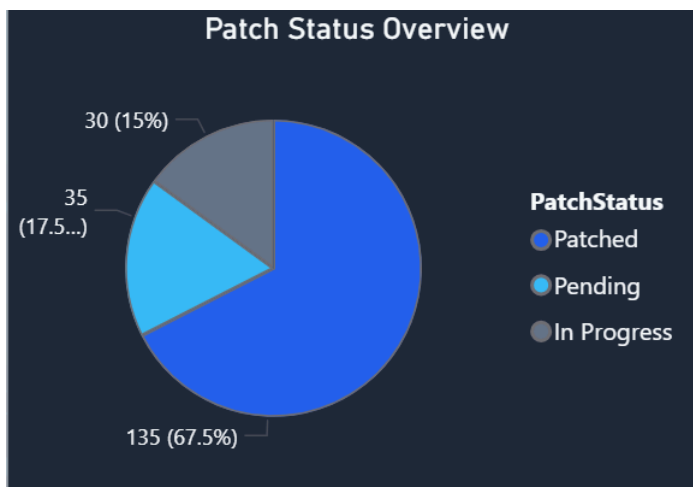
Blocked Attacks by Attack Type

A bar chart displaying the most common attack types, such as phishing, malware, ransomware, brute force attacks, and DDoS attacks.



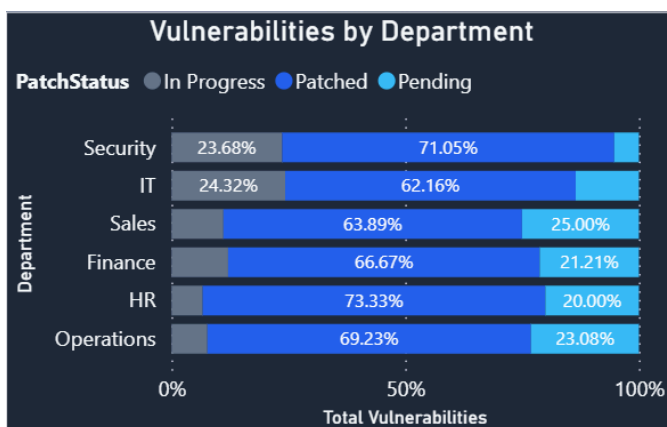
Global Attack Sources

A column chart showing attack origins by country, helping identify geographical attack patterns and support threat intelligence activities.



Patch Status Overview

A pie chart showing the distribution of vulnerabilities across different remediation stages, including Patched, In Progress, and Pending.



Vulnerabilities by Department

A stacked bar chart displaying vulnerability status across departments, helping identify areas requiring additional remediation efforts.

6. Key Findings

Threat Monitoring

A significant number of threat alerts were detected during the reporting period, indicating continuous monitoring requirements.

Attack Analysis

Phishing, malware, and DDoS attacks represent major attack vectors targeting the organization. A high attack block rate demonstrates the effectiveness of existing security controls.

Vulnerability Management

Most vulnerabilities have been patched successfully; however, some remain in progress or pending, highlighting opportunities to improve remediation processes.

Department-Level Insights

Certain departments show higher vulnerability counts than others, suggesting a need for focused security assessments and patch management activities.

7. Recommendations

1. Continuously monitor critical alerts and prioritize high-severity incidents.
2. Strengthen defenses against frequently occurring attack types such as phishing and malware.
3. Improve patch compliance by reducing pending vulnerabilities.
4. Conduct regular vulnerability assessments across all departments.
5. Monitor attack trends to identify emerging threats and adjust security controls accordingly.
6. Maintain proactive security monitoring to improve attack detection and prevention capabilities.

8. Benefits of the Dashboard

- Centralized cybersecurity monitoring
- Improved threat visibility
- Faster incident prioritization
- Enhanced attack analysis
- Better vulnerability management
- Improved decision-making through data visualization
- Interactive filtering and drill-down capabilities

9. Final Dashboard Look:



10. Report Page:



11. Conclusion

The Cyber Security Dashboard provides an integrated solution for monitoring threats, analyzing attacks, and tracking vulnerability remediation activities. By combining security metrics, trend analysis, and vulnerability management insights into a single platform, the dashboard supports effective cybersecurity operations and helps organizations strengthen their overall security posture.

This project demonstrates the use of Power BI for cybersecurity analytics, data visualization, KPI tracking, and security reporting.